

Rethinking Engineering Education

The CDIO Approach

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CHAPTER TEN

HISTORICAL ACCOUNTS OF ENGINEERING EDUCATION

U. JØRGENSEN

INTRODUCTION

When engaging in the reform of engineering education, it is important to understand its historical context. For over 150 years, educational institutions have played a major role in shaping the skills and professional identities of engineers. During this period, the appropriate approach to engineering education has been the subject of constant discussions and controversy. Major changes have occurred both in the way engineering education is organized and in its relation to science education. Radical changes have also occurred in the technologies and technical specialties within engineering. Despite this history, and particularly in view of the controversies surrounding the role of engineering education since the late 1960s, engineering schools have been surprisingly stable in their basic philosophy regarding the structure and core content of the engineering curriculum. Only modest reforms have been implemented in the curriculum and pedagogy of engineering education in several decades. Most of these reforms have been focused on increasing the number of technical engineering topics, and solving the resulting problems of disciplinary congestion.

By the 1990s, organized efforts in both the United States and Europe raised basic questions about the relevance of engineering education as it had developed since World War II. The problems included a lack of practical skills in modern engineering training, the lack of relevance for industry of the science being taught, and the kind of analytical qualifications being awarded in engineering education compared with visions of engineers as creative designers and innovators of future technologies. With its emphasis on science and knowledge structured around technical disciplines, engineering education developed into an education of technically skilled cooperative workers. However, many feel that the knowledge and broad innovative capacity needed to produce creative design engineers able to cope with contemporary technological change seem to be lacking in engineering education.

Several educational initiatives have addressed these issues, and attempted to outline plans to reform engineering education. Some focus on engineering curriculum or pedagogy; some develop completely new engineering programs based on new technologies. Other initiatives combine business, management,

and organizational understanding with engineering, or alternatively emphasize the creative and design aspects of engineering. Some reform initiatives have been supported by grants from government agencies, such as the National Science Foundation (NSF) in the United States; others have arisen from the Bologna Process that attempts to promote a unified system of education across Europe. While most initiatives focus on local, regional, or national experiments and reform, the CDIO Initiative is multinational, with open-source resources, and a broad, comprehensive methodology.

Contemporary tensions in engineering education may be deeply rooted in the diversity of modern technologies. The applications of these diverse technologies throughout society require increasing differentiation in the education of engineers. This diversity has already presented new challenges to the definition of engineering competence. The diversity of technologies presents new challenges to an engineering institution's sense of unity, identity, and standardization of professional preparation. Despite the complexity and multiplicity of technologies, institutional unity and its manifestation in a common engineering core curriculum have so far been successfully maintained by the engineering profession and by elite engineering universities. Nevertheless, the policies of identity formation and the creation of a homogeneous image of engineering are issues that need to be taken seriously, both in historical accounts and in contemporary reform initiatives. Engineering identity plays a vital part in educational reform and negotiations for change.

Critical accounts by observers close to the situation point to the need for reform in engineering education [1]–[2]. Other critics seem more confident in the achievements of engineers in society, and argue for the continuation of a traditional science-based engineering curriculum [3]. From their perspective, technology and the natural sciences are two distinctly separate approaches to knowledge [4]. Their studies contradict the popular misleading notion that engineering science is *applied science*. However, they do not raise critical issues related to the social and institutional dependencies of technology. Unfortunately, even engineering schools and professional institutions have supported the idea of a close relationship between science and technology by asserting that natural sciences are the core foundation of engineering. Contemporary developments in the natural sciences and engineering sciences have blurred the boundaries. New approaches of *techno-science* seem to be gaining ground as the characterization of the ties between modern science and technology, leaving neither one in a subsidiary role [5]. These new approaches recognize the role of technology as a contributor to scientific achievements, and change the basic idea of nature and technology.

The basic question is whether the critics are pointing to problems that will require radical reforms and transformations, or to a crisis in engineering education that will go away, as has happened so often before when technology and engineering have been criticized. The view that technology drives change and innovation seems to be less criticized today compared with the 1970s. At

the same time, there is a crisis in engineering practice, itself, that relates to problems in the conception and use of technology, and in the needs expressed by industry and society pushing for reform.

The objective of this chapter is to set the CDIO Initiative in a historical context that traces the tensions of engineering practices, institutional changes, identity formation, and technological developments that are the context for modern engineering education. The intent is to highlight the complexities of the historical context, and not necessarily to produce a natural evolutionary history of engineering education that led to the development of the Initiative. The first section describes the early establishment of civil (non-military) engineering education, and illustrates models of engineering education that reflected diverse national identities and perceptions of the role of engineers in society. The next section delineates the role of engineering in industrial and societal development and how that role created the framework for the classical engineering specialties that led to international standardization of engineering subjects. The third section emphasizes the transformation to a *science* of engineering after World War II when more engineering science subjects were added to traditional natural science subjects. The discussions highlight the move away from practical skills in recent decades due to the diminishing need for skilled technicians and craftsmen who once formed an important recruitment base for engineering education. A description of the contemporary explosion in the number of new engineering disciplines, schools, and programs focusing on technological domains leads to a discussion of the controversy about what should be the engineering core curriculum of the future.

CHAPTER OBJECTIVES

This chapter is designed so that you can

- recognize historic changes and institutional differences in engineering education
- recognize the controversy over how engineering experience and practice should be represented in education
- understand the contribution of engineering education to the construction of engineering identity
- explain the reasons that reform initiatives have developed in recent years.
- evaluate the controversy between engineering problem solving approaches and the natural sciences as the foundation for an engineering core curriculum
- be inspired to experiment with new ways to provide students with engineering competencies

THE GENESIS OF ENGINEERING EDUCATION

Tensions between theory and practice have permeated engineering education since its formal inception in the 19th century. Scholars in the United States have used the metaphor of a swinging pendulum to describe various waves of practical orientation versus theoretical priorities setting the agenda for engineering education [6]. A closer study reveals a spectrum of positions, ranging from practical, skilled, craft-based technical education to science-based educations that developed in engineering schools and technical universities. The idea of the swinging pendulum is also seen in institutions with extreme polar differences in identities and focus, as was the case for a long time in several European countries.

Engineering professions emerged during the 19th century. Civil engineering developed first, as an offshoot of military engineering, which focused on the construction of armaments, fortifications, and infrastructure [3]. Early industrial work was based on practical skills and crafts, and led to the establishment of technical schools. Engineering, on the other hand, was based on a vision of technical development and the use of systematic, analytical approaches, similar to the French idea of *polytechnique* [7]. This idea was developed and promoted through the building of *École Polytechnique* in 1792, marking the beginning of a new era of civil engineering education. The ideas permeated both Europe and the United States in the first half of the 19th century and led to the establishment of a new type of institution of higher education. At the same time, military schools, such as West Point in the U.S., were heavily influenced by the analytical approaches that developed from the polytechnic idea. The practical and theoretical approaches led to distinct institutional structures of technical engineering and engineering education in Europe and the U.S. What today may seem to be a homogeneous profession with a well-defined international identity, has a hard-won and conflict-ridden history. We now take a brief look at the evolution of engineering education in France, northern Europe, the United Kingdom, and the United States.

Engineering education in France

In France, engineering institutions developed according to the structure of French government institutions and industry [8]. Inspired by the idea of *polytechnique*, the *grande écoles* have been the core of French state education, setting the ideal standards for education of engineers. Besides working in government institutions, engineers were involved in creating the new infrastructures demanded by growing cities and industries in need of transportation, energy, and communication. In this context, the technical sciences were seen as applied sciences. This thinking was based on the assumption that mathematical theory and general principles of science would form a foundation on which to improve technology moving it from the level of practice and skill-based experience to a higher form of practical knowledge.

Other engineering schools were established, several of which were focused on emerging sectors of industrial importance, such as the mining and mechanical industries which supplied agriculture and factories with new technology-based equipment. Though practical training was included and the demands of industry influenced the content of the curriculum, the elitist structure of these engineering schools maintained the hierarchy and roles of theoretical training.

Engineering education in northern Europe

In northern Europe, the dominant structure of engineering education consists of two models of engineering recruitment and education. One model, the *fachhochschulen*, is based on a practical education that recruits skilled craftsmen from industry and trades. This line of education developed in the late 19th century from technical schools to supplement the skills of workers coming from apprenticeship-based craft training by providing more theoretical subjects ranging from technical drawing to calculus [9]. The second model is a university-like academic engineering education, typically differentiated from the more discipline-oriented university education in natural science. Often named *technische hochschulen* (renamed *technische universitäten* in the late 20th century), a variety of technical universities developed in Germany and the Scandinavian countries to meet local institutional traditions. The basis for these two models was the strong tradition of skilled workers and the continued differentiation in identity, supported by two different recruitment paths: one offered to the practically skilled engineers; the other to the academically trained engineers coming directly from secondary school.

The second model gained legitimacy from the idea that the technical universities contributed to the production of reason, while the model of the *fachhochschulen* established its legitimacy by emphasizing its contribution to progress through its focus on quality techniques, the usefulness of practical engineering skills in industry, and the application of technology [10]. Engineers with academic educations contributed for more than 50 years to the construction of societal infrastructures and institutions [11]. Some of these theoretically trained engineers contributed to the rise of new industries following inventions in chemistry and electronics. However, in the 19th century, the numbers of practically skilled engineers still dominated industrial development in both the mechanical industries and in mining. Even in Germany where the theoretical training at engineering universities was initiated and supported by the creation of research and development facilities in larger corporations, the contribution of engineers in industrial innovation came from their practical experiences and systematic experiments, and only in small part from theoretical, science-based knowledge [12].

Engineering education in the United Kingdom

In the United Kingdom, a quite different institutional model developed. Engineering was seen as growing from the practical, skilled crafts and was therefore kept from the universities and the sciences. Although the idea of a polytechnic education found its way to the U.K. in the form of polytechnic institutions, its implementation resembled the class structure in society where leadership in government and industry was dominated by university graduates, and where engineering was seen as a secondary trade—important but based in practical skills. This division kept engineering education at a distance from the universities for quite some time.

In addition to the specific character of engineering education and the image of practical work in the U.K., the British system of accreditation formed an important difference between the systems created in Germany and France, which dominated continental Europe. In Europe, government committees defined the qualifications of engineers through their educational programs. The British system of accreditation emphasized practical skills and engineering experience, and it also supported the idea that engineering competencies were of a different nature than the academic qualifications given by universities. The U.K. system of accreditation has, to some extent, been copied in the United States.

Engineering education in the United States

In the United States, mechanical engineering was one of the first fields of engineering focus together with civil engineering. It emerged from the rich and diverse machine shop and agricultural machine cultures that sprang up to support industrialization. The earliest institution, Rensselaer Polytechnic Institute was founded in 1824 and acquired its modern name in 1861. Although its name resembles *polytechnique*, Rensselaer exemplified an American approach to engineering education that emphasized practical, industrial, and agricultural experiences for students, with comparatively less emphasis on mathematics and science. Other schools founded in succeeding decades emulated this essentially advanced apprenticeship model. In the mid-19th century, the establishment of the *Agricultural and Mechanics* (A&M) schools and *land grant* schools, including the Massachusetts Institute of Technology in 1861, reinforced this practical approach with close ties to industry, a dominant focus on practical knowledge, machine shop work, with little independent faculty research.

During the late 19th century, American engineering educators, for example, Robert Thurston, recognized the strengths of the European systems, and began to advocate for an increased presence of science and mathematics in the curriculum. This view coincided with an increasing desire for professional respect for engineering equal to fields such as medicine and law. Thurston's agenda also included adding a research emphasis. Many activities were

included under this research umbrella, often through *engineering experiment stations* modeled on those in agriculture [6].

Although stirrings of new approaches appeared during the 1920s and 1930s, American engineering education remained largely within this practical, industrial orientation until World War II. In contrast, European schools, with such leaders as Felix Klein at Göttingen, excelled at applying scientific and theoretical approaches to engineering problems. During this period, intellectual leaders in the U.S., for example, Theodore von Kármán (a student of Klein), with European educations [6], transferred the idea of a more science-based type of engineering training to American institutions.

The practical approach of engineering institutions in the U.S. and in the polytechnics and *fachhochschulens* in Europe were of major importance to the development and implementation of technology in industry and society. These institutions influenced the formation of a professional engineering identity. Although this fact is recognized in contemporary discussions, it is overshadowed by the focus on the theoretical, science-based training that forms the modern ideal of formalized engineering teaching. The tension originates from the creation of an engineering identity where attempts to distance engineers from skilled technicians and their apprentice-based training resulted in a focus on an academic tradition based on the vision of the polytechnic institutions of higher education.

ENGINEERING AND INDUSTRIAL DEVELOPMENT

Many engineering universities and schools originate from civil and mechanical engineering developed during the first part of the 19th century. Their graduates were employed in government institutions, or involved in the creation of public and private companies that developed with the building of new infrastructures: transportation systems, roads, bridges, harbors, canals, ships, sewers, water supplies, and eventually, systems producing and distributing gas. Engineers' responsibilities and contributions to progress were based on their roles as constructors of the material pillars of modern society. Later, the view of their roles expanded to include engineers as innovators and system builders because of their contributions to new institutions, new knowledge, and the technical infrastructure [2]. To legitimize large investments in infrastructure, decision makers required hard data, and this need matched the focus on formal, science-based knowledge. The need to legitimize development also supported the creation of hierarchical and bureaucratic technological institutions. The idea of a technocracy that could support and even contribute to government policy was, therefore, consistent with the basic patterns and structures of knowledge created in relation to these large construction and infrastructure projects. For example, the connection in France between the idea of a *polytechnique* and the role of government bureaucracy

is illustrated in the ideas of Hans Christian Oerstedt, the Danish founder of the *Polyteknisk Lærestalt* (now the Technical University of Denmark) in 1829. Oerstedt saw a close relationship between polytechnic education and education in the political sciences in the German and Danish *staatswissenschaft* [13].

The role of military organizations, and the inspiration from large infrastructure projects employing military engineers also had a great influence on the factory systems established in large corporations [14]-[15]. These corporate systems were inspired by the military model of hierarchical organization and the need for unity and standards. The quest for standards also gave rise to the idea of improving productivity and sustaining control over the production process and work force through the use of scientific management principles, which soon became a core element of engineering management.

The tensions between the polytechnic training based on physics and mathematics versus practical skills in technical drawing and laboratory experiments were evident from the beginning. The controversies were fueled by the work of new polytechnic graduates, trained in the natural sciences, but primarily engaged in constructing the new technical infrastructures, that is, the water systems, sewers, gas pipes, and later, electrical power systems. The polytechnic graduates were also involved in establishing transport infrastructures, such as canals and rails, and the new communication infrastructures, such as telegraph, telephone and radio systems connecting cities, regions and nations.

In contrast, many of the machine shop developments and early industrial achievements were as much the result of the practically skilled technicians and craftsmen's work. They based their knowledge on the experiences of working with the construction of machines and chemical processes in industry, and they transferred their knowledge through visits to other sites. Traveling, working abroad, and returning with knowledge of technical constructions and innovations, detailed technical drawings and descriptions of new machines were common ways of transferring knowledge and new technologies [16]. The diffusion of the new technical constructs was supported by journals, most often national, such as the Danish *Polyteknisk Tidsskrift*, edited by Ursin, the first professor in mechanics at the *Polyteknisk Lærestalt*.

During early industrial development, technical schools supplied many of the inventors of new machines, tools, and production systems. Practically skilled engineers, recruited from these schools, played an important part in industrialization until the late 19th century [11]. These engineers were involved in the new industries because of their experience with rationally organized experiments, documentation processes, and their experience with the achievements of developing technologies [17]. Though engineering disciplines show similar patterns across countries, differences of engineering practice exist across national cultures, especially in the way the theoretical contributions from engineering science are used in engineering practice [18]. In the later part of the 19th century, when research and innovations in

petrochemicals gave a boost to the chemical industries and energy distribution systems, academic research and academically trained engineers gained importance. In this way, the developments of petrochemical and electrical technologies led to changes in the role of technical institutions of higher education, particularly in northern Europe.

The structure of many engineering institutions who built their curriculums on the *big four* in engineering – civil, mechanical, chemical and electrical – originates from this period. Although engineering schools still tended to train their students to solve practical industrial problems, and academic research was often difficult to distinguish from industrial consulting, electrical engineering was the exception. In this engineering discipline, the relationship between theoretical teaching and industrially developed technologies was closer than in other engineering domains. Still, many universities maintained basic engineering skills by requiring electrical engineers to study mechanics, technical drawing, and surveying. These requirements could not be explained in relation to the knowledge and skills needed in the new field of electrical engineering, but were established as part of the early curriculum for engineering education, in which mechanical and civil engineering practices were the standard.

In the course of history, many engineering disciplines developed from what could be called an *encyclopedia stage*, dominated by descriptive representations of technological exemplars, into a more abstracted and theory-based *scientific stage* [19]-[20]. This latter stage adds the strength of applying model descriptions, including mathematical representations and topic generalizations. However, in the transformation process, concrete experiences and practice-based knowledge, embedded in specific technical solutions, were often lost. Consequently, the transition represents a movement from scattered collections of representational exemplars to more complete representations of the technologies in question, documented by constructed theories and models. At the same time, the transition represents movement away from the engineering practice and experiences that are needed to make technology functional [21].

SCIENCE AS THE BASIS FOR ENGINEERING

In order to understand today's situation, we must consider one of the most important historical changes in engineering education – the construction of a science base for engineering. This development resulted from the increase in public and military funding of engineering research during World War II. The program to establish a science base for engineering created a new *elite* of theoretically oriented universities and technical schools of higher education in both the United States and Europe. At the outset, there was a gap in engineering curricula between science classes based on high degrees of mathematically formalized knowledge, and the more descriptive and less codified technical subjects. Earlier controversies resulted in positioning technical sciences as

secondary, or applied, in relation to the natural sciences. Technical universities, at least in Europe, were restricted from giving doctoral degrees and addressing scientific matters without the support of university faculty versed in the natural sciences. However, the new era of expanding technical sciences lessened these controversies because of its increased focus on innovation and awareness of the close interactions between specific areas of science and technology.

Developments in the United States

The watershed event in American engineering was World War II. One of the leading institutions in this change was The Massachusetts Institute of Technology. Before the war, MIT had embraced scientific approaches, under the presidency of physicist Karl Compton. Vannevar Bush, a young MIT faculty member, reoriented his research from circuit simulations for electric power networks to general research in calculating machines – a more scientific orientation – successfully attracting private foundation support [22]. In 1940, Bush set up the National Defense Research Council, a major federal wartime research establishment in Washington, DC. Although engineers made significant contributions during the war, the success of the Manhattan Project put physicists in the spotlight, and savvy engineering leaders recognized that the path to prestige lay in engineers' closer emulation of scientists.

Developments in Europe

In Europe, this orientation toward a scientific basis for engineering already had a long tradition in the intellectual environment around the elite institutions, especially in France and Germany. The post-war tendency toward formalization of science councils and large government-sponsored research programs, centered on the peaceful utilization of technologies developed during World War II, spurred a dramatic increase in research at technical universities, and a change in the methods of teaching engineering. During the first half of the 19th century, several natural science subjects, for example in Germany and Denmark, were taught either in common at the universities and the polytechnic institutions, or only at the latter, so that natural sciences students had to take lectures at the polytechnic institutions. When the natural sciences became established within the traditional universities, they increasingly were perceived as being the foundation for the applied sciences.

During the first half of the 20th century, polytechnic universities had to fight for acceptance. They were acknowledged for their foundations in science, but were questioned about whether they could conduct independent scientific research; or were limited to practical experiments with technical improvements and practical implementation. These controversies manifested themselves in the acceptance of doctoral studies at technical schools of higher education. In Sweden and Germany, as in many other countries, decisions about what should qualify as scientific achievement and who was

qualified to judge were very controversial. The controversy ended with an acceptance of technical or engineering science as a distinct area of scientific inquiry, although the image of engineering science as *merely* applied natural science continued to dominate many discussions about the character and role of technical sciences. Some engineering schools began granting degrees in engineering science to their best students [24].

Post-war developments

The movement toward a science base was concurrent with a massive post-war expansion of government-funded research in the United States. Sponsorship of fundamental studies in a variety of areas supported the trend away from practice-oriented research and education. Successes in fields such as high-speed aerodynamics, semiconductor electronics, and computing confirmed that physics and mathematics, conducted in a laboratory-based environment, could open new technological frontiers. Military research during these years also tended to focus on performance – increased power, higher altitudes, faster speeds – goals that were conducive to scientific approaches.

Electrical engineering, for example, no longer focused on electric power and rotating machinery, but instead, on electronics, communications theory, and computing machines. As historian Bruce Seely [6] wrote:

Theoretical studies counted for much more than practice-oriented testing projects; published papers and grants replaced patents and industrial experience as measures of good faculty. By the mid-1960s, the transition to an analytical and more scientific style was largely completed at most American engineering colleges."

Yet today, many engineering departments still have their core activities defined by technical disciplines, such as mechanics, energy systems, electronics, chemistry, building construction, or sanitary and civil engineering. Many of these disciplines have specific problems and industries that relate to their founding years, but as the demand for science-based research and teaching became prominent, the original roots to practice and industry lost their significance. With the changing demands, more abstract courses, and courses defined by scientific fields, were developed.

The post-war decades saw the rise of systems engineering and thinking as broadly applicable engineering tools [22]. Systems sciences that include control theory, systems theory, systems engineering, operations research, systems dynamics, cybernetics and others led engineers to concentrate on building analytical models of small-scale and large-scale systems, often making use of the new tools provided by digital computers and simulations [25]. Techniques range from practical managerial tools, such as systems engineering, to technical formalisms, such as control theory, to more mathematical formulations, such as operations research. A broad-based movement within engineering found that these tools might finally provide the theoretical basis for all engineering that goes beyond the basic principles provided by the natural

sciences. Whereas systems engineering of the 1950s could be narrowly analytical and hierarchically organized, new ideas of systems in the 1980s and 1990s focused on the relationship between technology and its social and industrial context. This new relationship and understanding of the natural and technical sciences is reflected in the notion that engineering as technoscience developed in the field of sociological studies of science and technology to reflect the new intimate relationship between these fields of science [26].

THE DECREASE IN PRACTICAL SKILLS AND EXPERIENCE

The creation of the research university as the ideal and elite model for engineering universities also influenced the staffing of engineering education's lecturer positions. The increase in research-based funding of these positions meant that the tradition of hiring practitioners to lecture in engineering was increasingly supplemented or replaced by lecturers hired on the basis of their achievements in engineering science and laboratory work, instead of their achievements in industrial practice. Voices were raised both inside and outside the universities against this change, resulting in the transformation of almost all lecturer and academic research staff at technical universities. However, within the universities, most objections and arguments came from practitioners who were involved in teaching and laboratory work identified as routine and trivial in comparison with frontline research.

The transformation of technical schools

In the European setting, the requirement of a Ph.D. degree narrows the recruitment of engineering faculty, making it difficult for engineers with careers in industry to satisfy entry requirements for university professorships. New Ph.D. recipients have been entering, in increasing numbers, into research positions funded by government programs, with fewer going to industrial laboratories and engineering practice. Even though the requirement of a Ph.D. degree can be substituted by personal innovative activities in industry, it is difficult to recruit practically skilled engineers to universities. Academic positions, today, require applicants to document research activities and demonstrate published works from their research in order to be evaluated for appointment. This threshold, in combination with a gap in the wage levels of industry compared with universities, has reduced the number of qualified engineers with skilled engineering practice in universities.

Changes in the foundation of engineering education, with the expansion of science-based technical disciplines, also led to changes in the curriculum of traditional vocational schools of engineering, as well as funding for research. Though with different names, the *polytechnics* in the United Kingdom, the

fachhochschulen in Germany, and the *teknika* in Denmark shared common characteristics in recruiting students from groups of skilled technicians and supplementing their training with a theoretical education, while maintaining a focus on industrial practice. As a result, the schools inherited the experience-based, practical knowledge, and skills of students who had previously worked as apprentices in construction firms, machine shops, and industry. During the 1960s, the curriculum of these technical schools was expanded, and many of their specialized lines of engineering education were extended in length and scope. Typically, these changes included improvements in mathematics and natural sciences by copying the science base from engineering universities, while attempting to maintain their practical orientation. This led to the appointment of government committees to address the profile of practical engineering education [27]. It also raised questions about the balance between the academics and practice, and whether these schools would continue to supply practice-based engineers to industry.

At the same time, the decline in the apprenticeship training of craftsmen and skilled workers began to undermine the recruitment lines of the polytechnics [28]. While this type of engineering education was well supplied by the traditional, smaller crafts-based industries, the growth in the size of industries led to a change in the ways the workforce was trained, leading to an increasingly specialized machine shop skills in the workforce. Fewer candidates had the necessary broad skills and apprenticeship training required by the engineering schools. The schools were forced to establish other recruitment systems to survive. This process resulted in a complete reversal of the basis for recruiting students during the 1990s. As a result, it is difficult today to distinguish the two different lines of engineering education from one another, both because of the convergence of their student enrollments and the nature of their educational focus.

The response from industry

The response from industry to the tensions in technical education demonstrated the ambiguity of industry's interests in maintaining practically skilled engineers. Industry was not willing to carry the costs of an educational system to maintain the basic skills needed in the workforce. More generally, the problem also demonstrated the ambiguities in understanding which aspects of practice and experience were important for engineering work. Studies of engineering have demonstrated the importance of combining formal theoretical work, based on codified knowledge, with methods of drawing, experimentation, models, and analogous reasoning [16]. These skills cannot be based solely on the practical experience of shop-floor technicians, but also on the experiences of practicing engineers. Other practical perspectives, such as having experienced the daily routines of industrial organizations can be gained from other practices than working as an engineer. The ability of engineering institutions to recruit students with practical skills may have

diminished, but the problem of maintaining the practical aspects of engineering competence continues to exist [29].

The return to practice

During the 1970s, a variety of technical and political events began to change the course of technology in its social context, and began to swing the pendulum yet again toward practice. The oil shocks, the beginning of the modern environmental movement, and the cancellation of the supersonic transport (SST) in the United States were indicators that technology might no longer progress along strictly technical lines. During the 1980s, the U.S. found itself in a crisis of competitiveness, which some blamed on the engineering research establishment's excessive focus on performance and military research, as opposed to other more industrial considerations. The *Made in America* study at MIT [30] reported that design and manufacturing had not received the academic resources or the intellectual prestige of the engineering sciences, and hence the U.S. had fallen behind such rivals as Germany and Japan in actually producing consumer goods. At the same time, the end of the cold war meant that large, military-oriented research funds might no longer be forthcoming. During the 1990s, academic institutions increasingly turned toward industrial sources of support. Along with new sources of money came new research orientations toward product design, product development, and innovation studies, with more emphasis on problems from engineering practice.

From within the technical universities, voices were raised against the consequences of a too-narrow focus on science-based teaching that lacked interest in the practical aspects of engineering work and competence [31]. Educational programs focusing on project work and problem-based learning, introduced in some experimental engineering education programs during the 1970s, spread broadly during the 1990s. They attempted to address the problems from a pedagogical and didactic point of view. In both Denmark and Germany, a few radical reform universities made project-oriented study the trademark of their education, stating that the projects could both cater to the interdisciplinary aspects of engineering methods and problem solving, and to the integration of the practical and theoretical elements needed in engineering [32].

While shop-floor training and practical aspects of work organization were the focus in the earlier phases of engineering, the new perspectives on engineering practice emphasize the complexity of engineering tasks, including project organization and communication, the role of specialized consultants, the skills needed to handle innovative design tasks, and the need to include the social dimensions of technology [33]. These new emphases may not eliminate the need for practical skills in drawing, visualization, modeling, and crafting of material objects, but the replication of traditional crafts does not satisfy the need for practical training in engineering. New emphases create a need to redefine *engineering practice* and to leave the apprenticeship model behind.

DISCIPLINARY CONGESTION AND BLURRING BOUNDARIES

The growth of the use of technology in the later half of the 20th century, in combination with the large investments made in engineering research by industry and by research institutes and universities, has resulted in tremendous growth in the body of technological knowledge, the number of new technological domains, and specialized technical science disciplines [34]. Differentiation in engineering specialties put pressure on engineering education to cope with the diversity and to keep up with the frontline of knowledge in the diverse fields. At many institutions, this resulted in a number of new specializations. Several of these specializations relate to sectors and industries that, for shorter or longer periods of time, have required engineers with particular kinds of knowledge. Changes in the demands for specialization created tension between generalized engineering knowledge and the specialized knowledge needed in individual domains of technology and engineering practice. Examples of these specializations include highway engineering, ship building, sanitary engineering, mining engineering, power generation and distribution engineering, offshore engineering, aeronautics, microcircuit engineering, environmental engineering, bio-engineering, multimedia engineering, and wind turbine engineering.

Alternatives for addressing disciplinary congestion

All these specializations led to an expansion in the numbers and variety of courses focusing on technical sciences. At some technical universities, for example, the Massachusetts Institute of Technology and the Technical University of Denmark (DTU), the curriculum was organized into modules, giving students choices about how to structure their own education. While some universities expanded the number of specializations, others coped with disciplinary congestion through negotiation of the core content and opted for elective courses in only a limited part of the curriculum. General pedagogical reform based on project-oriented work also argued for giving students a broad understanding of engineering work and problem solving, with less emphasis on theoretical knowledge represented in the courses and disciplines. A very different response had been to question the concept of engineering education altogether, by giving more space in the curriculum to science-based teaching, by reducing the number of laboratory classes, and by weakening the ties to industry and the technological domains from which engineering originated.

Blurring boundaries between technology and nature

The dominant role of technology demands multidisciplinary approaches, and challenges the science-based, rational models and problem-solving approaches. These demands gave rise to new areas of engineering education. For example,

in the field of environmental studies, the need for new approaches in industry based on cleaner technologies and product chain management challenged the already established disciplines in sanitary engineering based on end-of-pipe technologies and chemical analysis. From focusing on nature as a recipient of wastes, engineers had to realize that nature itself has been dramatically affected, and that environmental knowledge had to include the design of production processes and chemicals as part of what had become a continued re-design of nature. Blurring boundaries between technology and nature had introduced serious ethical and political issues into the core of engineering.

Another example can be found in the field of housing and building construction engineering. The need for integrating both social and aesthetic elements, as well as user interaction in both the project and use phases of construction, led to several attempts to overcome the traditional division between civil engineering and architecture. Several engineering education departments tried to solve this problem by employing staff from different disciplines – engineers, architects, and sociologists – hoping that solutions would emerge from the multidisciplinary melting pot. In several cases, the integration turned out difficult; housing construction and city planning in engineering crumbled in spite of the attempts. This dilemma left engineering housing construction departments in situations where the focus became more theoretical rather than contributing to the design and functionality of building construction. In contrast, functionality, usability, and flexibility, as well as the inclusion of users in the planning of building design, were left to the architects, who seemed more interested in aesthetics. This example illustrates the dominance of disciplinary culture in engineering schools, and the ways in which that culture defines and constructs new strands of knowledge and scientific research.

The influence of new technologies

Changes in the role of technologies in a society where consumer uses, complex production, and infrastructures are increasingly more important, have led to more focus on the integration of usability and design features. The traditional jobs in processing and production have not vanished, but new jobs in consulting, design, and marketing have been created. These new jobs demand new personal and professional competencies, and require new disciplines that contribute to the knowledge base [35]-[36]. During the 1990s, several engineering schools started new lines of education emphasizing engineering design skills and introduced aspects of social sciences into the curriculum of engineering design. These additions included technology studies, user ethnographies, and market analysis. The development of new and diverse technologies also reflects the limitations of technical sciences in being able to cover all aspects of engineering [31]-[37]. Examples of these reformed engineering programs can be found at Delft University in the Netherlands, Rensselaer Polytechnic Institute in the U.S., the Technical University of Denmark, the Norwegian University of Science and Technology, and Cranfield University in the U.K.

The growth and diversity of technological knowledge also leaves universities with continued pressure for renewal and difficulty in determining which engineering domains to maintain and develop. Several domains and branches of industry have passed their growth phase, and the related technologies are no longer the focus of research funding. The given industry may still be employing large numbers of engineers, but its need for new engineers does not justify creating or sustaining programs in engineering schools. For engineering education, it means the potential loss of important domains of technological knowledge.

The decade of the 1990s was not the first time that concerns about the role of technology in society had surfaced, but this time the questions raised issues of a more fundamental nature concerning the content of engineering education and the impact on technology exemplified with controversies about highway planning, chemicals in agriculture, nuclear power plants, and the social impacts of automation. The concerns questioned the role of knowledge about technology and some critics demanded a humanistic input into the curriculum with such subjects as ethics, history, philosophy, and disciplines from the social sciences [38]. This idea was based on the assumption that engineering students, through confrontation with alternate positions and opportunities to discuss social and ethical issues, would be better prepared to meet the challenges of technology. However, in most engineering education programs, these new subjects ended up being add-on disciplines often not integrated with engineering and science subjects, contributing further to the disciplinary congestion in engineering [39].

Developments in technology also have meant that the boundaries between engineering disciplines are blurring, and indeed the very nature and existence of engineering has come into question in recent years. What used to be fairly distinct areas of engineering – civil, mechanical, chemical, electrical – have now become combinations of two or more fields and their disciplines. For example, there are now programs in civil and environment engineering, aeronautics and astronautics, electrical engineering and computer science, and materials science and engineering. New programs in bioengineering and biomaterials reflect these shifts as well [1]. Today, many of the larger technical universities offer programs in more than a dozen different engineering fields.

Technological change has changed the face of engineering in many other ways, as well. Engineering research and design are changing, due, in no small part, to computers and the Internet. Algorithms, once taught as fundamental skills, are now built into automated design software. Large projects are run and coordinated through digital links between people who may never have met face to face. What had been largely a white male profession is now diverse in race, national origin, and gender. In addition, it is now possible for companies to conduct engineering functions worldwide with the help of automation and new technologies. The idea of well-defined boundaries for engineering education has been challenged by new technological domains and by existing university educations that already address technology as part of the curriculum.

Areas that address technology and have close affiliations with engineering represent a broad variety of subjects and approaches, for example, pharmaceuticals, architecture, computer science, information technology, environmental studies, biotechnology, nanotechnology, and technology management. These professional areas do not necessarily see themselves as part of engineering. In some areas, new perspectives of techno-science can create new relationships between science and technology. New fields of biotechnology and nanotechnology have blurred the boundaries with the natural sciences, as well, leading to the creation of such fields as mathematical engineering and nanotechnology in the natural sciences. Thus, a situation is developing where the new professionals, industrialists, and politicians question whether technology remains the domain solely of engineers, and whether engineering will continue to be the major source and producer of innovation. This development has been called *expansive disintegration* [1], reflecting the combined expansion of the number of technologies, specialties and disciplines on the one hand, and the continued disintegration of what once was the unity and identity of engineering on the other. These transformations will fundamentally challenge the role of engineering schools in the future.

CONTEMPORARY CHALLENGES

The role of engineers in technology and innovation is often taken for granted. Even in future-oriented reports on engineering, there is a tendency to expect problem solving abilities in societal and environmental issues from engineering, without challenging contemporary foundations of engineering curricula [23]. New insights coming from innovation theory, demonstrating a broader scope in innovation, coupled with changes in the societal use of technology that imply growing complexity and a need for social skills, point to the need for improvement in engineering education. On the other hand, innovations during the last decade are leading to changes that may make the role of engineering less central in the future. Policy and management attempts to govern innovation processes have also broadened the scope and shifted the focus from technological development and breakthroughs to a broader focus on market demands, strategic issues, and the use of technologies.

A new identity for engineering

Early in the 20th century, the idea that engineers have societal responsibility and are the heroic constructors of the material structures of modern society was being supplanted by a less heroic and more mundane image of engineers as the *servants* of industry. This image of engineering reflects a reduction in the influence of engineers on the direction and content of technological innovation, and supports the positioning of engineers in a less influential and subordinate role in their attempts to promote business interests [13]-[36].

This view is not all that different from engineers' self-image in contemporary society. The description of an engineer's competencies might include the following: possesses a scientific base of engineering knowledge, problem-solving capabilities, and the ability to adapt their knowledge and practices to new types of problems. The focus is more often on problem solving, and less on problem identification and definition [37]. This focus emphasizes the problem of engineering identity in distinguishing between engineers as creators and designers versus analysts and scientists. Although engineers' identity as creators and designers is supported in historical writing and in strategic reports about the role of engineering in the future [23], reality seems to place engineers in roles closer to analysts and scientists in laboratories and modern technical industries.

The underlying assumption in the discussion about engineering problem solving is that engineers are working with well-defined technical problems and methods from an existing number of engineering disciplines. This assumption does not answer the question as to whether engineers are competent in handling non-standardized social and technical processes where the problems are undefined and involve new ways of combining knowledge. Simply broadening the science base in a more interdisciplinary direction, including the social sciences and humanities, may not have been a satisfactory solution. The mere addition of topics to the curriculum does not change engineering practices or provide a better integration of knowledge [1]. A new engineering identity will be based on the answers to these questions:

- *What competencies are necessary to manage the creative, socio-technical and design skills that need to be improved in engineering education?*
- *What is the meaning of engineering problem identification and problem solving today, and how can they be reflected in engineering education?*

A new education for engineers

The reforms in engineering education, initiated in the 1970s in some engineering schools, emphasized the need for problem solving and project work that simulated real engineering practice, but these reforms did not provide the complete answer. The response lies in a new understanding of the role of science in innovation and the use of technology in context. This approach underlines the existing need to bridge the divide between the disciplinary knowledge of the technical sciences and social sciences, and the practical domains of engineering, with their unique knowledge and routines that integrate the social, practical, and technical aspects of technology at work [24]. It is necessary to rethink disciplinary knowledge as presented in engineering education, and a corresponding need to reform the content and structure of that knowledge.

One solution might be to accept that the idea of a single unifying engineering identity has proven to be problematic. Engineering education will unavoidably become more diverse in the future. Integrating engineering into the general university structure [1] could be a tempting solution, removing

the rigid focus on core curriculum, while still fighting the battle for the acceptance of engineering science. However, the problems of including professional, practical knowledge and maintaining the need for professional skills in engineering are not solved by referring students to an even more diverse science base at universities. Neither do the many new science-based specializations in engineering provide a solution, and may even bring engineering further away from the practical knowledge also needed. Most technical disciplines focus on particular technical solutions taught as individual courses and with less emphasis on their application. These courses are supposed to contribute to a coherent set of engineering competencies, although they have little resemblance to an established domain of engineering practical problem solving and solutions [40].

Debates on engineering education tend to replicate a number of discussions over and over again. One example is the balance between practical skills and theoretical knowledge. While the debate may seem the same, the content has changed radically during the more than one century of controversy [41]-[42]. The list of relevant practical skills would not be the same, and similarly, theoretical knowledge has evolved as a result of technological developments, advanced tools, computers, and simulation models. Reforms need to produce a new realization of the kind of practical insights relevant to engineering education today.

Another unsettled challenge involves the balance between specialist and generalist knowledge in engineering. A process occurs in which the *current* deep knowledge and skills continuously changes. New knowledge and skills that begin as part of a science frontier are considered demanding. As the frontline of technological innovation moves, that knowledge and those skills become part of standard engineering procedures, technical standards, standardized components, design concepts, and are supported by computerized tools and simulation models. What counts as core or basic disciplines in an engineering curriculum changes in the wake of the expansion of new engineering domains and disciplines, despite the fact that all are dominated by the idea of a common theoretical foundation.

A new education for engineers will answer these questions:

- *What content should a core engineering curriculum have in the future?*
- *Which skills should be part of the curriculum, and which can be developed on the job after completing the education?*
- *What is the sequence of knowledge from abstract knowledge to practical application?*

Based on the insights from didactics, it is important to realize that the idea of a ‘natural order of knowledge’ starting in the most abstract and general disciplines and ending with application does not provide a good answer to curriculum planning. Instead, a critical approach to the role of knowledge in learning and the creation of engineering identity may be needed to overcome the taken-for-granted approaches in curriculum planning. Reform in

engineering education needs to address the contemporary challenges highlighted here and find new ways of analyzing and understanding technological knowledge and the professional practices of engineering.

Addressing contemporary challenges with a CDIO approach

From the historical accounts presented in this chapter, we find a number of challenges that continue to be important in contemporary debates about the reform of engineering education. Key elements of the CDIO approach reflect critical issues underscored in this historical account. These elements include:

- A curriculum that includes personal and interpersonal skills, and product, process, and system building skills in the realization of technical solutions, while still emphasizing mathematics, the natural sciences, engineering science, and the technical knowledge specific to technological domains
- A core belief that personal and interpersonal skills, and product, process, and system building skills must be learned in the context of authentic problem solving and engineering practice
- An integrated approach to the teaching of engineering science and engineering disciplinary knowledge relevant to specific technologies and engineering practice

The CDIO Initiative renews the focus on several of the important issues in engineering educational reform that have been part of its history. While many reform initiatives have focused on the curriculum of different science and engineering disciplines, the balance between practical education and theory-based learning, or the role of project-based learning, the Initiative has created an approach that coordinates all the elements required to address the challenges of engineering in a modern, complex, technological society.

SUMMARY

In this chapter, we have shown that the engineering profession and engineering education have developed differently in European countries and the United States, demonstrating national differences in the role of engineering in society and industry and the ways engineering education has been influenced. The military use of infrastructure and equipment and the ingenuity of practically skilled builders pointed to the roots and inspiration for the creation of formal training of engineers for civil purposes and industrial development. Later developments emphasized the role of science in engineering training. The relationship between scientific theoretical knowledge in the disciplines, and the practical skills and knowledge derived from technological innovation and engineering practice has been controversial in engineering education and is still presents challenges for contemporary engineering education.

The differences among countries has led to the implementation of different educational structures characterized both by the background students were expected to have before entering the programs, and the ways in which lecturing was combined with laboratory and machine shop work. In the United States, with one dominant model of engineering education, the changing balance between practical and theoretical education has been characterized as a swinging pendulum, while in Europe, typically, two models of engineering institutions were created with differing weights placed on the two aspects of engineering.

With the rise of engineering science, especially since World War II, the core elements of engineering have been extended beyond the natural sciences. At the same time, the growth in the number of scientific specialties led to a focus on theoretical, science-based engineering education, leaving aside some of the creative design aspects of engineering, as well as some of the experiences and routines from the domains of engineering practice.

Engineering education has faced problems of a growing number of specializations with a consequent disciplinary congestion in the curriculum. At the same time, new areas of technology and new professional specializations outside engineering schools have introduced topics that are difficult to distinguish from engineering, blurring the boundaries between the professions. This blurring of boundaries, together with changes in the image of engineers away from *creator* toward *technical industrial worker*, have challenged the identity and content of engineering for the future.

The chapter concludes with questions about the knowledge base for engineering in modern society and its influence on a core curriculum for engineering education. This chapter points to a need to re-assess the composition of engineering education concerning the way practical skills and experiences are combined with theoretical training and the core elements of an engineering curriculum.

DISCUSSION QUESTIONS

1. Can you identify traces of the controversies in the history of engineering education, presented in this chapter, at your institution or in your professional society?
2. How would you characterize engineering education in your country? Can you identify one dominant type of engineering program in the way it balances theory and practice? Is there more than one model?
3. Do you see any limitations or barriers to an engineering program based solely on science and engineering disciplines? How are social issues of technology integrated into the curriculum?
4. In what ways can the CDIO approach change the trajectory of engineering education reform?

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